

**UNIVERSITY OF CATANIA**  
**DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE**  
**MASTER'S DEGREE IN DATA SCIENCE**

# **DEEP LEARNING FOR SILENT LIP READING PROJECT PRESENTATION**

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# CONTEXT AND MOTIVATION

Silent Lip Reading aims to decode human speech by analyzing visual signal from lip movements, completely excluding audio information. This technology helps in noisy places, for security cameras (CCTV) and to help people with hearing disabilities.

**Core challenge:** The main obstacle is that some words sound different but look the same on the lips. These are called "Visemes."

## PROJECT OBJECTIVES

- Analysis of a dataset cropped on lips (no audio);
- Training Neural Networks to classify 76 words;
- Comparison of data splitting strategies.

# Exploratory Data Analysis

## INITIAL STATE

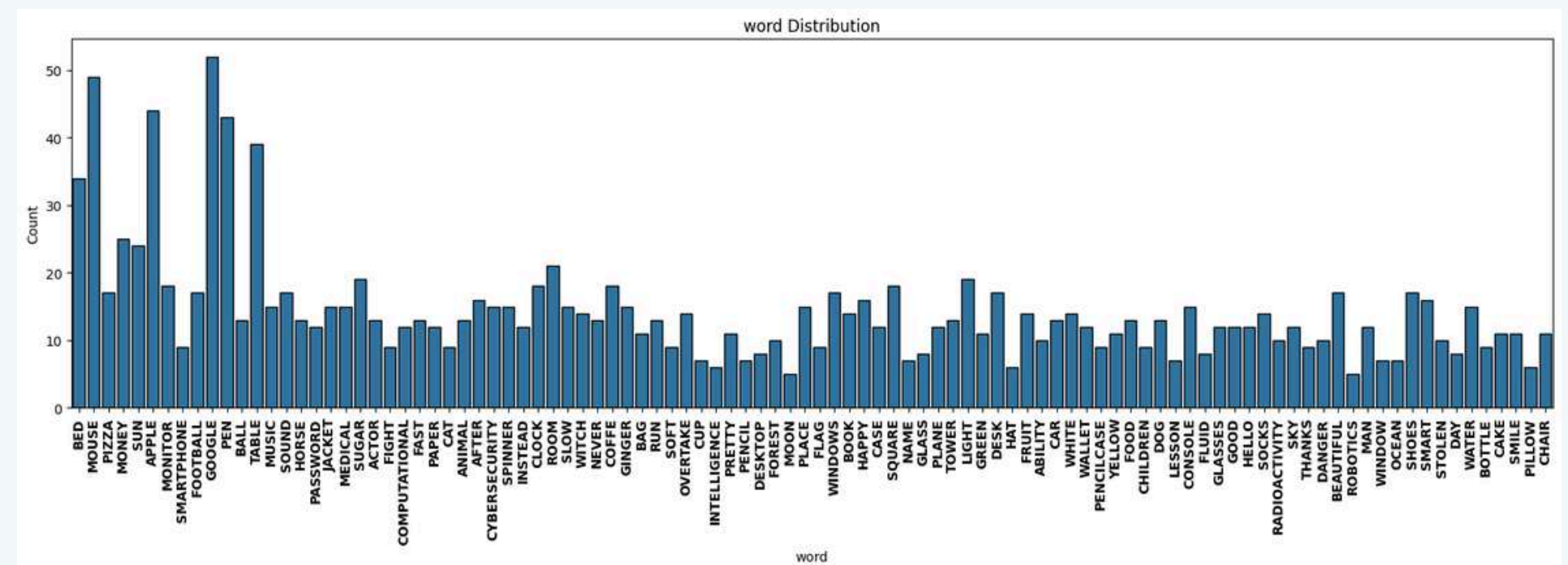
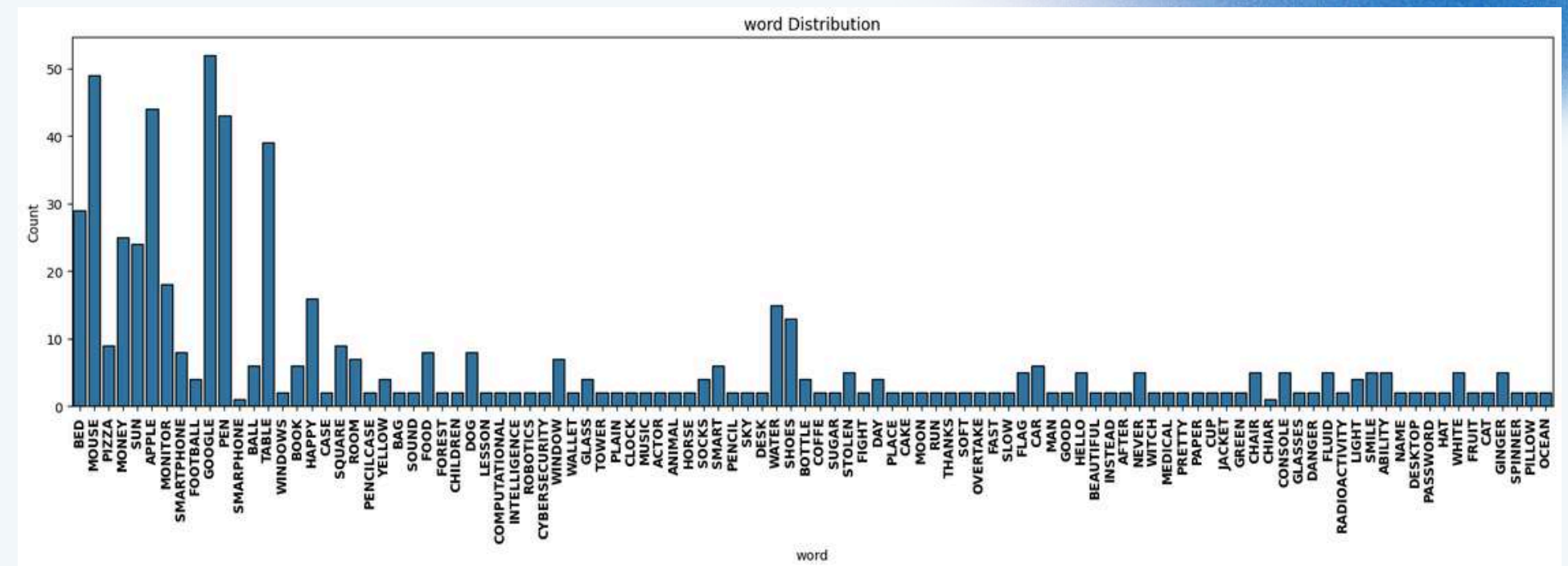
Our analysis started with a dataset of **650 videos**. The EDA revealed a **severe class imbalance in word frequency** and **insufficient samples to train deep networks** effectively.

## DATA ACQUISITION AND FINAL STRUCTURE

To address this scarcity issue, we performed a **data acquisition campaign** with university colleagues.

We **recorded new videos, cropped the Region of Interest around the lips**, and **standardized the resolution** to 256 x 128 pixels.

**This expanded our dataset to 1,439 samples** with 76 active classes (words).



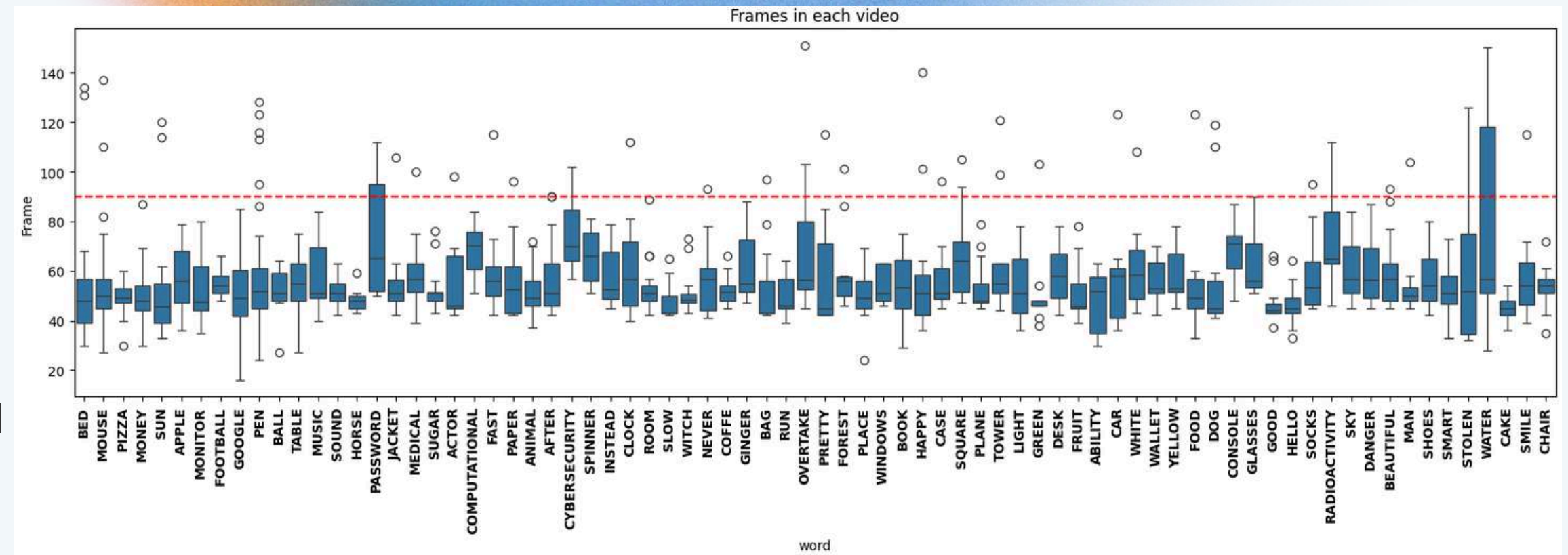


# TEMPORAL ANALYSIS

## DURATION ANALYSIS

While the maximum duration reaches 150 frames, **the interquartile range shows that the vast majority of speech sequences are significantly shorter.**

We selected **90 frames** as the **optimal trade-off**, covering approximately **95% of the dataset.**



## NORMALIZATION STRATEGY

Videos to the left of the line are **Zero-Padded to reach 90 frames**, while outliers to the right are **Truncated** to remove non-informative tail frames. This step ensures that the Neural Network receives a **consistent input tensor**, optimizing the **balance between information retention and computational cost.**



# NEURAL ARCHITECTURES

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## 01 LOGISTIC REGRESSION

This **baseline approach** is used to establish a **minimum performance benchmark**.

## 03 HYBRID (CNN + LSTM)

An Hybrid architecture **combining CNNs** for spatial feature extraction and **LSTMs** for **temporal** modeling

## 02 2D-CNN

Applied frame-by-frame (**Spatial** analysis only).

## 04 TRANSFER LEARNING

MobileNetV2 trained on **ImageNet** + LSTM

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## FINAL MODEL

Our best model utilizes **Transfer Learning with a MobileNetV2** backbone **pre-trained on ImageNet**, providing robust visual features that significantly improve the network's ability **to "see" lip shapes**.

**LSTM** is used to understand the **sequence of words**.

**To prevent overfitting**, we implemented dynamic callbacks such as **EarlyStopping** and **ReduceLROnPlateau** to adjust the learning rate during training.

# EXPERIMENTAL RESULTS

The performance of the models was heavily influenced by the **data splitting strategy**. We identified three distinct scenarios.

1. **Scenario A: Open Set (Subject Independent)**: The model is tested on people it has never seen before.
2. **Scenario B: Mixed Split**: The model knows the person, but has never seen them say that specific word (it learned the word from others).
3. **Scenario C: Closed Set (Subject Dependent)**: The model knows the person and has seen them pronounce that word in other videos.



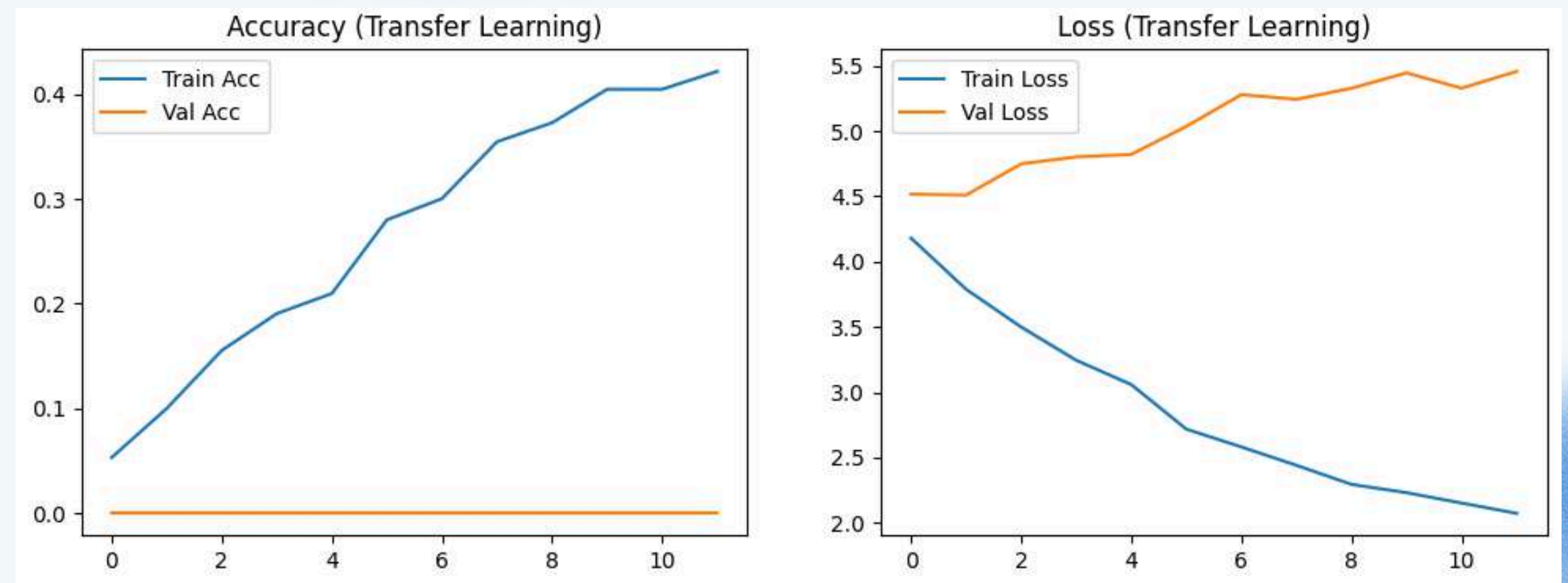
# SCENARIO A – OPEN SET

## PROTOCOL

In this scenario we evaluated the model's ability to generalize by testing it on users who were never seen during the training phase.

## RESULT

The model achieved an accuracy between 0% and 2%, indicating a complete failure to generalize to new faces.



## INTERPRETATION

This result highlights a significant "Subject Gap," where the model learns to recognize the somatic features of the speaker rather than the universal lip dynamics.



# SCENARIO B – MIXED STRATEGY

## PROTOCOL

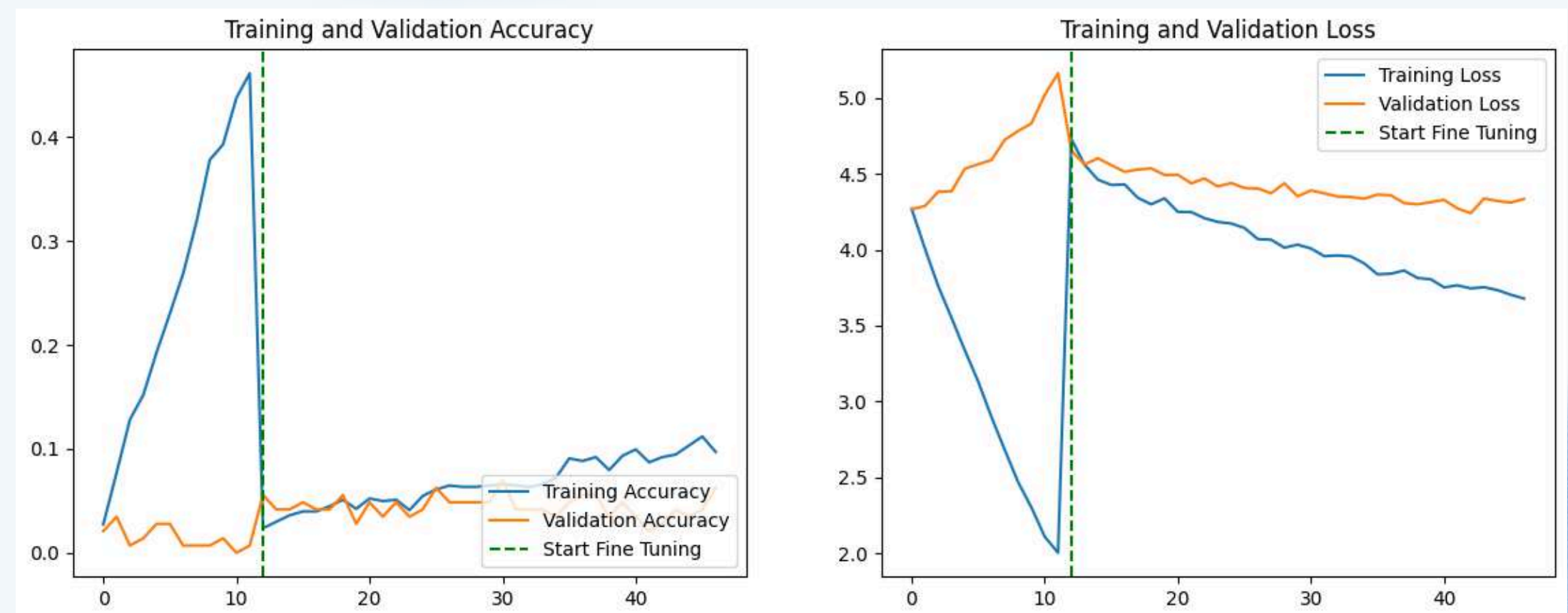
We attempted to facilitate the task by including known users in the Test Set, but evaluating them on specific word instances that were pronounced by other users in the Training Set.

## RESULT

The accuracy improved only marginally to approximately 3%, which remains insufficient for any practical application or real-world deployment.

## INTERPRETATION

This suggests that simply knowing the face is not enough; the high variability in how different people pronounce the same word prevents the model from generalizing without more data.





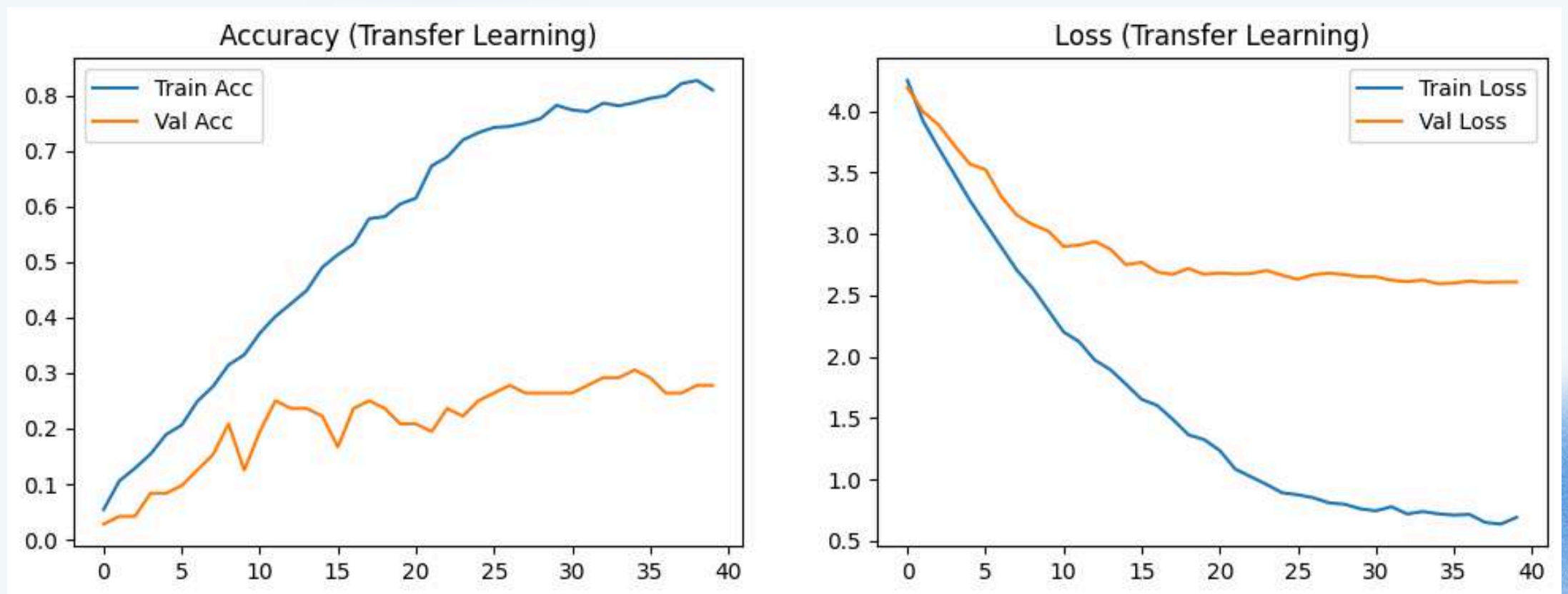
# SCENARIO C – CLOSED SET

## PROTOCOL

Finally, we shifted to a Subject Dependent approach, ensuring that the Training and Test sets shared the exact same pool of users (evaluating on different video instances).

## RESULT

The model achieved a significant breakthrough with an accuracy of approximately 30%, vastly outperforming all previous subject-independent attempts.



## INTERPRETATION

This proves that the MobileNetV2 architecture is valid and capable of learning lip dynamics, provided it has been trained on the specific speaker's facial traits.



# EXPERIMENTAL RESULTS SUMMARY

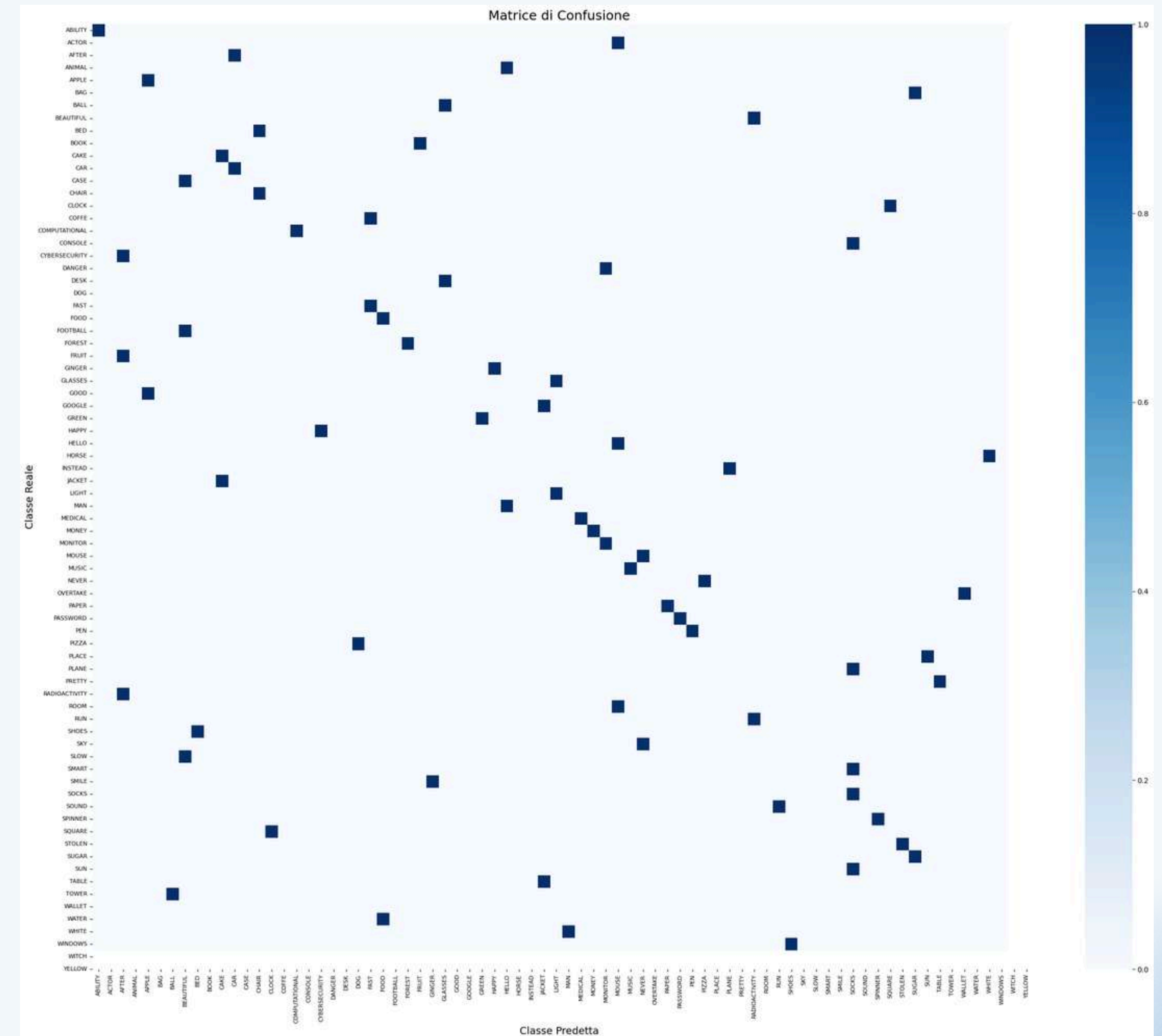
| SPLITTING DATASET | ARCHITECTURE        | TRAIN ACCURACY | TEST ACCURACY | OUTCOME                           |
|-------------------|---------------------|----------------|---------------|-----------------------------------|
| OPEN SET          | LOGISTIC REGRESSION | ~ 6.5 %        | ~ 2%          | Insufficient complexity.          |
| OPEN SET          | 2D-CNN              | ~ 6.5 %        | ~ 0%          | Fails without temporal memory.    |
| OPEN SET          | 2D-CNN + LSTM       | ~ 40 %         | ~ 2%          | It doesn't recognize unkown users |
| OPEN SET          | MOBILENETV2 + LSTM  | ~ 40 %         | ~ 0%          | It doesn't recognize unkown users |
| CLOSED SET        | MOBILENETV2 + LSTM  | ~ 79 %         | ~ 30 %        | Good performance with known users |

# ERROR ANALYSIS: CONFUSION MATRIX

Despite reaching 30% Validation Accuracy in the Closed Set scenario, the performance indicates room for improvement. We employed diagnostic tools to understand the remaining errors.

The **Confusion Matrix** shows a **validation accuracy of approximately 30% along the main diagonal**. **Misclassifications are widely distributed across the matrix.**

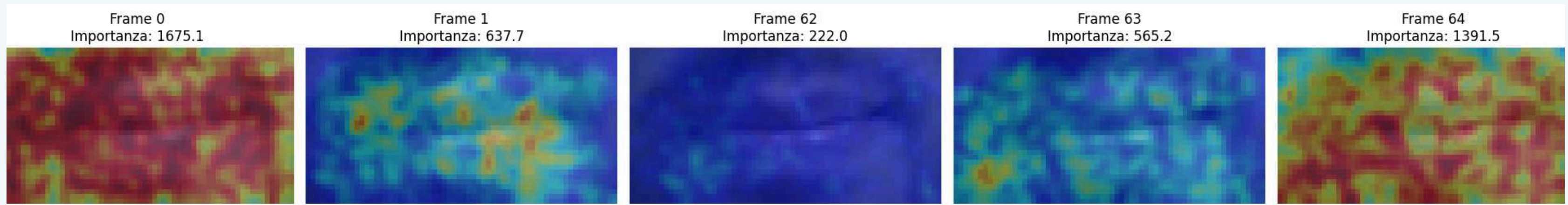
This dispersion suggests that the **primary challenge** derives from the **high dimensionality of the classification problem** (76 classes), which **significantly increases the complexity of distinguishing between target words** compared to a smaller dataset.





# ERROR ANALYSIS: GRAD CAM

To investigate the failure of the poor performance, we used **Gradient-weighted Class Activation Mapping (GradCAM)** to visualize the network's focus.



The heatmaps revealed that the model frequently focused on irrelevant somatic features, such as the chin, beard, or background, rather than the lips.

This provides visual evidence that the model was overfitting: it identified who was speaking based on static traits, rather than decoding what was being said.

# CONCLUSIONS

This project highlighted the complexity of Silent Lip Reading. Through the transition from a standard CNN to a Transfer Learning approach with MobileNetV2 and LSTM, we established that:

- **Data Quantity is Critical:** A dataset of ~1,200 videos is insufficient for Open Set generalization (Subject Independent), resulting in near-zero validation accuracy.
- **Feasibility in Closed Sets:** In a Subject Dependent context, the model successfully learns to classify words with 30% accuracy, a promising baseline given the high number of classes (76).
- **Future Work:** To bridge the gap between Open Set and Closed Set performances, future developments should focus on massively increasing the dataset size and exploring 3D CNNs to better capture spatiotemporal features.



**THANKS FOR  
THE ATTENTION**